Spectral type can vary within a limit-periodic Schrödinger hull

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3 November 2026

Abstract

We construct a bounded real limit-periodic potential whose uniform hull contains both an operator with an explicit square-summable eigenfunction and an operator with no eigenvalues. The two phases have different nonzero spectral components, answering AIM 1018 negatively. The construction also permits arbitrarily small potentials.

Repository target. This manuscript concerns AIM problem 1018 at revision `3fa50884d129`. Its scope is stated below.

Status

This is an elementary counterexample to the exact catalogue statement of problem 1018, namely Damanik–Fillman Question 10.2 [1, arXiv v1, p. 36]; see also [2, Remark (v)]. Every construction and estimate needed below is included. The construction gives an explicit phase with a positive, square-summable eigenfunction at energy zero, and an explicit phase in the same uniform hull with no eigenvalues at all. Consequently their sets of nonzero spectral components differ, whether or not the first operator has any additional spectral component.

1. Periodic triangular functions on an odometer

Let $G = \mathbb{Z}_2$ be the additive group of 2-adic integers, and write $x \mapsto x + 1$ for its minimal translation. Set

$$P_k = 16^k, \qquad \epsilon_k = 2^{-k}, \qquad k \geq 1,$$

On the cyclic group $\mathbb{Z}/P_k\mathbb{Z}$, use its usual nearest-neighbor distance d_k. Define the real, locally constant function

$$h_k(x) = \epsilon_k \, d_k(x \bmod P_k, 0). \tag{1}$$

These functions satisfy:

1. $h_k \geq 0$, $h_k(-x) = h_k(x)$, and $h_k(0) = 0$.
2. h_k has period P_k under integer translation.
3. $|h_k(x + 1) - h_k(x)| \leq \epsilon_k$ for every $x \in G$.

Put

$$b_k(x) = h_k(x+1) - h_k(x), \qquad b(x) = \sum_{k=1}^{\infty} b_k(x). \tag{2}$$

Because $\sum \epsilon_k = 1$, this series converges uniformly to a continuous real function with $\|b\|_\infty \le 1$. Moreover,

$$b(-x-1) = -b(x). \tag{3}$$

The functions h_k themselves need not have a convergent sum on all of G; no such claim is used.

2. A bounded, even sampling function

Define

$$F(x) = -e^{-b(x)} - e^{b(x-1)}, \qquad V_x(n) = F(x+n), \quad n \in \mathbb{Z}. \tag{4}$$

Then F is real and continuous, $\|F\|_\infty \le 2e$, and (3) gives

$$F(-x) = -e^{-b(-x)} - e^{b(-x-1)} = -e^{b(x-1)} - e^{-b(x)} = F(x). \tag{5}$$

Let $b^{(K)} = \sum_{k=1}^{K} b_k$ and $F^{(K)}(x) = -e^{-b^{(K)}(x)} - e^{b^{(K)}(x-1)}$. Each $F^{(K)}$ is even and P_K-periodic. The mean value theorem gives the uniform bound

$$\|F - F^{(K)}\|_\infty \le 2e\, 2^{-K}. \tag{6}$$

Thus every V_x is a bounded uniform limit of periodic sequences, exactly as required in the problem.

The map $x \mapsto V_x$ is continuous from G to $\ell^\infty(\mathbb{Z})$: this follows either from uniform continuity of F in the translation-invariant group metric, or directly from (6) and finite-level periodicity. Integer translations are dense in G, so

$$\Omega_{V_0} = \{V_x : x \in G\}. \tag{7}$$

Injectivity of $x \mapsto V_x$ is unnecessary.

3. An explicit eigenfunction at one phase

For each integer n, define

$$H(n) = \sum_{k=1}^{\infty} h_k(n). \tag{8}$$

The series converges to a finite value: by the Lipschitz bound and $h_k(0) = 0$,

$$0 \le h_k(n) \le \epsilon_k |n|, \qquad 0 \le H(n) \le |n|. \tag{9}$$

Whenever $P_k \ge 2|n|$, the cyclic distance from n to zero equals $|n|$, so

$$h_k(n) = \epsilon_k |n|. \tag{10}$$

For $|n| \geq 1$, choose the least $K \geq 1$ with $16^K \geq 2|n|$. Then $16^K \leq 32|n|$, including the case $K = 1$. Therefore

$$H(n) \geq |n| \sum_{k=K}^{\infty} 2^{-k} = 2|n|\, 2^{-K} \geq \frac{2}{32^{1/4}} |n|^{3/4}. \tag{11}$$

Consequently

$$u(n) = e^{-H(n)} \tag{12}$$

is positive and belongs to $\ell^2(\mathbb{Z})$. It is nonzero because $u(0) = 1$. Both H and u are even sequences. The absolutely convergent series in (8) and uniformly convergent series in (2) show

$$H(n+1) - H(n) = b(n). \tag{13}$$

Thus

$$\frac{u(n+1)}{u(n)} = e^{-b(n)}, \qquad \frac{u(n-1)}{u(n)} = e^{b(n-1)}.$$

Substituting (4) gives the exact eigenvalue equation

$$u(n-1) + u(n+1) + V_0(n)u(n) = 0. \tag{14}$$

Hence 0 is an eigenvalue of H_{V_0}, and $\mathrm{pp} \in \mathcal{T}(V_0)$.

For context, this eigenvalue is at the upper spectral edge. For every finitely supported sequence f and every phase x,

$$-\langle f, H_{V_x} f \rangle = \sum_{n \in \mathbb{Z}} e^{b(x+n)} \left| f(n+1) - e^{-b(x+n)} f(n) \right|^2 \geq 0.$$

This follows by expansion; boundedness extends it to all of ℓ^2. Thus every operator is nonpositive, while the phase zero operator has the displayed nonzero kernel vector. The potential V_0 is necessarily nonperiodic: a full-line periodic second-order difference operator has no square-summable eigenfunction, as follows directly by iterating its fixed, invertible period transfer matrix in both directions.

4. An explicit phase with arbitrarily accurate distant reflections

Set $K_1 = 1$. Recursively define

$$n_j = \sum_{r=1}^{j} P_{K_r}, \qquad K_{j+1} = n_j^2 + K_j + 1. \tag{15}$$

The sequence K_j is strictly increasing, and

$$\theta_* = -\sum_{r=1}^{\infty} P_{K_r} \tag{16}$$

converges in G. For each j, the element $\theta_* + n_j$ is divisible by $P_{K_{j+1}}$. By evenness and periodicity of $F^{(K_{j+1})}$,

$$F^{(K_{j+1})}(\theta_* + 2n_j - m) = F^{(K_{j+1})}(\theta_* + m), \qquad m \in \mathbb{Z}.$$

Writing $W = V_{\theta_*}$ and using (6), we get

$$d_j := \sup_{m\in\mathbb{Z}} |W(2n_j - m) - W(m)| \le 4e\, 2^{-K_{j+1}} \le 4e\, 2^{-n_j^2}. \tag{17}$$

In particular $n_j \to +\infty$ and $e^{Cn_j}\sqrt{d_j} \to 0$ for every fixed $C > 0$.

5. Elementary reflection lemma excluding all eigenvalues

Lemma. Let W be a bounded real sequence. Suppose there are positive integers $n_j \to \infty$ and nonnegative d_j satisfying

$$\sup_m |W(2n_j - m) - W(m)| \le d_j, \qquad e^{Cn_j}\sqrt{d_j} \longrightarrow 0$$

for every $C > 0$. Then H_W has no $\ell^2(\mathbb{Z})$ eigenfunction.

Proof. Suppose $H_W u = Eu$ with u real, nonzero, and $\|u\|_2 = 1$. Put $v_j(m) = u(2n_j - m)$ and

$$\mathcal{W}_j(m) = u(m+1)v_j(m) - u(m)v_j(m+1).$$

Subtracting the two difference equations yields

$$\mathcal{W}_j(m) - \mathcal{W}_j(m-1) = [W(2n_j - m) - W(m)]u(m)v_j(m).$$

Since both sequences tend to zero at infinity, summation from $-\infty$ and Cauchy–Schwarz give $|\mathcal{W}_j(m)| \le d_j$. At the reflection center,

$$|u(n_j)[u(n_j+1) - u(n_j-1)]| \le d_j. \tag{18}$$

If $|u(n_j)| \le \sqrt{d_j}$, set $w_j = u + v_j$. Then

$$w_j(n_j) = 2u(n_j), \qquad w_j(n_j+1) = (E - W(n_j))u(n_j),$$

so this two-component vector is $O_{E,\|W\|_\infty}(\sqrt{d_j})$. Otherwise set $w_j = u - v_j$. Then $w_j(n_j) = 0$ and (18) gives $|w_j(n_j+1)| \le \sqrt{d_j}$.

In either case, $(H_W - E)w_j$ has absolute value at most d_j at each site. The inverse one-step transfer matrices have norm at most a constant $C_E \ge 1$ depending only on E and $\|W\|_\infty$. Propagating the two-component vector backwards from n_j to zero gives

$$|w_j(0)| + |w_j(1)| \le C'_E C_E^{n_j}(\sqrt{d_j} + n_j d_j) \longrightarrow 0.$$

But $v_j(0) = u(2n_j) \to 0$ and $v_j(1) = u(2n_j - 1) \to 0$, so necessarily $u(0) = u(1) = 0$. The second-order recurrence then forces $u \equiv 0$, a contradiction. This also covers $d_j = 0$ by the same case distinction. □

Applying the lemma to (17) shows $\mathrm{pp} \notin \mathcal{T}(V_{\theta_*})$.

6. Conclusion, small potentials, and limits of the assertion

Both V_0 and V_{θ_*} lie in the same uniform limit-periodic hull by (7), but

$$\mathrm{pp} \in \mathcal{T}(V_0), \qquad \mathrm{pp} \notin \mathcal{T}(V_{\theta_*}).$$

Therefore the answer to the exact problem 1018 is **no**.

Both behaviors occur densely in the hull

The hull elements with a nonzero point component include the translation orbit of V_0, which is dense. The elements with no point spectrum contain a dense G_δ subset, as follows directly from the reflection lemma.

For an integer n, write $(R_n W)(m) = W(2n - m)$. In the compact metric hull $\Omega = \Omega_{V_0}$, define

$$\mathcal{U}_j = \bigcup_{n \geq j} \{W \in \Omega : \|W - R_n W\|_\infty < e^{-n^2}\}, \qquad j \geq 1.$$

These sets are open because each reflection map is an isometry in the uniform norm. Each V_{-n} is exactly symmetric about n, since F is even. The set $\{V_{-n} : n \geq j\}$ is dense in the hull, so $\mathcal{U}_j$ is dense. Baire's theorem makes $\bigcap_{j \geq 1} \mathcal{U}_j$ a dense G_δ. Every potential in that intersection has an unbounded sequence of reflection centers with errors at most e^{-n^2}, and hence has no eigenvalues by the lemma. Thus the two different point-spectrum behaviors both occur densely.

Arbitrarily small examples

The counterexamples can be made arbitrarily small after a constant energy shift. Indeed, replace ϵ_k by $\delta 2^{-k}$ with any $\delta > 0$. Then $\|b\|_\infty \leq \delta$, the lower bound (11) is multiplied by δ, and the eigenfunction remains square-summable. Moreover,

$$\|F + 2\|_\infty \leq 2(e^\delta - 1) \longrightarrow 0 \quad (\delta \downarrow 0).$$

Adding the constant 2 to every hull element simply shifts its spectrum by 2, preserving all spectral types. The reflection error bound becomes $4e^\delta \delta 2^{-K_{j+1}}$, which still satisfies the lemma. Thus hull spectral-type variation occurs for arbitrarily small uniform limit-periodic potentials.

The proof does not classify the rest of either operator's spectrum, does not show that H_{V_0} is pure point, and by itself does not settle the separate mixed-type question 1021 or the coupling question 1020. It is unnecessary to classify those remaining components to distinguish the two sets $\mathcal{T}$.

The construction is reproducible from the displayed formulas alone and uses no numerical evidence. The related classical theory of continuous cocycles with discrete orbits over odometers provides conceptual context, but is not invoked in the proof: J. Kwiatkowski and A. Siemaszko, *Discrete orbits in topologically transitive cylindrical transformations*, Discrete Contin. Dyn. Syst. **27** (2010), 945–961, https://doi.org/10.3934/dcds.2010.27.945.

References

[1] D. Damanik and J. Fillman, *Spectral Properties of Limit-Periodic Operators*, in *Analysis and Geometry on Graphs and Manifolds*, M. Keller, D. Lenz and R. K. Wojciechowski (eds.), London Mathematical Society Lecture Note Series 461, Cambridge University Press (2020),

382–444. doi:10.1017/9781108615259.016. The question and section locators used here refer to https://arxiv.org/abs/1802.05794v1.

[2] D. Damanik and Z. Gan, *Spectral properties of limit-periodic Schrödinger operators*, Communications on Pure and Applied Analysis **10** (2011), 859–871. doi:10.3934/cpaa.2011.10.859. https://arxiv.org/abs/0906.3337v3.

[3] J. Kwiatkowski and A. Siemaszko, *Discrete orbits in topologically transitive cylindrical transformations*, Discrete and Continuous Dynamical Systems **27** (2010), 945–961. doi:10.3934/dcds.2010.27.945. This reference supplies context for discrete cocycles, not a theorem used in the self-contained construction.